

Coffra Marketing Dashboard

A live, branded marketing analytics dashboard with transparent data provenance



Live Demo: coffra-marketing-dashboard.streamlit.app

Project	P2 · Coffra Marketing Analytics Dashboard
Author	Sebastian Kradyel
Date	April 2026
Repository	github.com/sebikradyel1-svg/coffra-marketing-automation
Live URL	coffra-marketing-dashboard.streamlit.app
Status	v1.0 — deployed to Streamlit Cloud
Stack	Streamlit · Plotly · pandas · HubSpot Private App API

Executive Summary

P2 builds on P1's strategic and operational foundation by delivering a live, deployable marketing analytics dashboard. The dashboard provides operational visibility across five domains: lead quality (XGBoost model insights), AI subject line optimization performance, HubSpot CRM snapshot data, simulated campaign funnel analytics, and a methodology page with full data provenance disclosure.

The deliverable solves a specific portfolio problem: most marketing portfolios are static documents (PDFs, presentations, screenshots). A recruiter scanning ten portfolios remembers the one with a clickable link to a working application. This dashboard is that link.

The technical approach prioritizes honesty over impressiveness. Rather than fabricating live campaign data for a fictional brand, the dashboard layers real data (model outputs, AI tool cache, HubSpot snapshot) with simulated funnel metrics that are clearly labeled and grounded in published industry benchmarks. The methodology page documents this approach explicitly.

Key Outcomes

Component	Outcome
Pages built	5 (Lead Quality, Subject Optimizer, HubSpot CRM, Campaign Funnel, Methodology)

Component	Outcome
Data sources integrated	3 real (model artifacts, optimizer cache, HubSpot snapshot) + 1 simulated (clearly labeled)
Deployment	Streamlit Cloud free tier with auto-rebuild on git push
Brand consistency	Light theme + Coffra brown across all pages, matching P1 case study and email design
Honest disclosure	Methodology page documents what is real, snapshot, simulated for every metric shown

Problem and Approach

The portfolio differentiation challenge

The AI Marketing Specialist job market in 2026 has converged on a familiar set of portfolio artifacts: persona documents, email copy samples, ML notebooks, and PDF case studies. These are necessary but no longer differentiating. Recruiters scanning GitHub repositories of qualified candidates increasingly look for live, deployable applications that demonstrate production readiness.

Strategic decisions

Decision	Choice	Rationale
Data strategy	Snapshot over live API	Stable demo post-trial-expiry; recruiter sees consistent state
Real vs simulated data	Mix with explicit labeling	Honesty signals senior judgment; fabrication signals junior overreach
Theme	Force light theme via config	Recruiter sees branded version regardless of system preference
Deployment	Streamlit Cloud free tier	Zero-cost public URL; auto-rebuild on git push
Pages structure	Multi-page sidebar nav	Mirrors enterprise dashboards; supports separate concerns per page

Architecture

The dashboard is a multi-page Streamlit application following standard separation of concerns. Shared utilities live in **dashboard/lib/** and individual pages live in **dashboard/pages/**, auto-discovered by Streamlit's page registry.

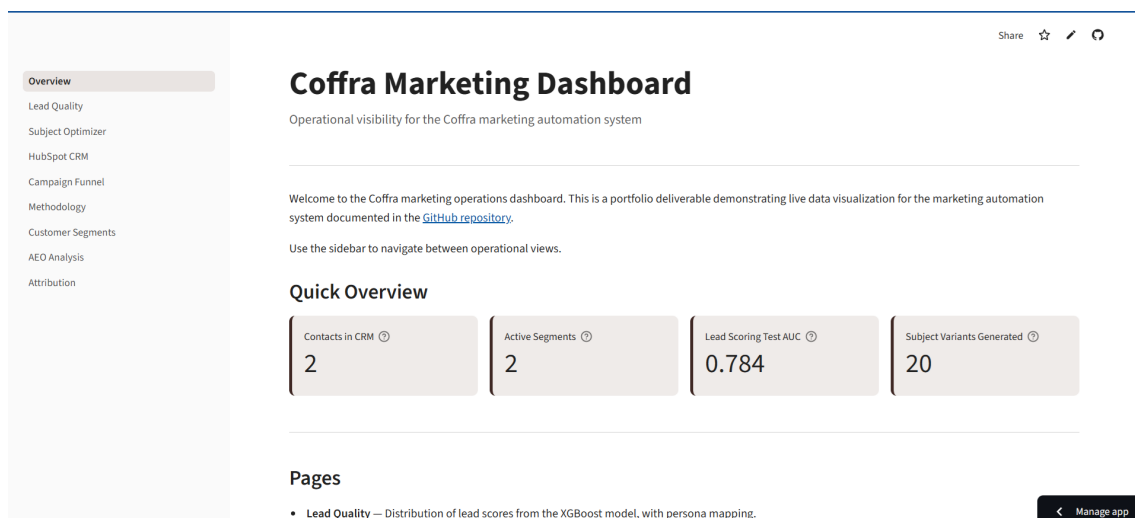
File structure

Path	Purpose
dashboard/streamlit_dashboard.py	Main entry point, overview page with cross-section metrics
dashboard/lib/styling.py	CSS injection, brand colors, page headers, data disclosure banners
dashboard/lib/data_loaders.py	Cached file I/O for snapshots, model artifacts, optimizer cache
dashboard/lib/plots.py	Brand-styled Plotly chart factories (bar, donut, funnel, gauge, line, histogram)
dashboard/pages/1_Lead_Quality.py	XGBoost model performance, baseline comparison, sample predictions
dashboard/pages/2_Subject_Optimizer.py	Cache analysis, persona coverage, character distribution
dashboard/pages/3_HubSpot_CRM.py	Contacts, segments, persona distribution from snapshot
dashboard/pages/4_Campaign_Funnel.py	Per-persona funnels, engagement decay, cart recovery (simulated)
dashboard/pages/5_Methodology.py	Data provenance disclosure, production roadmap
.streamlit/config.toml	Theme configuration (light theme, Coffra brand colors)
extract_hubspot_snapshot.py	One-time HubSpot Private App API extraction script

Data flow

All data is read from local files committed to the repository. The HubSpot snapshot is captured once via the extraction script and stored as JSON. The lead scoring model artifacts come from the P1 notebook output. The Subject Optimizer cache is generated during normal use of the P1 tool. No runtime API calls are made by the dashboard, ensuring it remains functional after the HubSpot trial expires.

This architecture is deliberately simple. A production system would add: scheduled snapshot refresh, fallback to live API on staleness, authentication, role-based access control, and a proper data warehouse layer. These extensions are documented in the Methodology page's production roadmap.



Pages Walkthrough

Lead Quality

Visualizes the XGBoost lead scoring model from P1. Top section shows ROC-AUC and PR-AUC as gauge charts for at-a-glance performance. Comparison table below contrasts Logistic Regression baseline vs XGBoost across cross-validation and held-out test metrics. The class imbalance disclosure (88% positive class — atypical) is shown explicitly with mitigations applied (class_weight, scale_pos_weight, PR-AUC reporting). Sample predictions table shows 20 test set rows mapped to Coffra lead segments (Cold / Warm MQL / Sales-Ready). Feature engineering section discloses dropped leakage columns and engineered features.

Subject Optimizer

Reads the local cache from the P1 Subject Line Optimizer (Claude API responses). Shows total generations, evaluations, variant count, and estimated USD cost based on token usage. Persona coverage donut indicates whether the tool has been exercised across both personas. Strategic angle distribution shows whether the prompt's 5 angles (direct, curiosity, insider, counter-intuitive, invitation) are balanced. Character length analysis validates mobile preview compliance (under 50 chars). Recent variants table and selected winners section provide qualitative review.

HubSpot CRM

Renders the snapshot extracted from HubSpot Marketing Hub trial via the Private App API. Includes contact count, segment count, persona distribution donut, and tables of individual contacts and segments. Persona internal values from HubSpot (persona_1, persona_2) are mapped to human-readable labels (Connoisseur, Daily Ritualist) in the loader. A workflows section documents the three workflows configured in HubSpot, with a note on the 'Send marketing email' lock-out and migration path. Raw snapshot JSON is available in an expander for transparency.

Campaign Funnel

The only page using simulated data, clearly labeled with an orange-red disclosure banner. Two parallel funnels (Connoisseur and Daily Ritualist) model a 14-day nurture campaign with 500 starting subscribers each. Conversion rates are anchored to Mailchimp 2025 Food & Beverage benchmarks (22.1% open rate, 2.3% CTR) with a 1.15x premium-brand uplift factor. Engagement decay charts show realistic open/click trajectories across the 5-email sequence. Cart recovery section models the 3-email recovery sequence with industry-standard recovery rates (15-25% range). All assumptions are documented inline.

Methodology

The transparency layer of the dashboard. Documents data provenance per page (real, snapshot, simulated), data age (when each source was captured and when it goes stale), what the dashboard does NOT include and why (no real campaign data, no fabricated ROI), and the production roadmap describing how to scale this into a live system. Designed to be the page a senior recruiter opens first to assess judgment.

Notable Technical Decisions

Snapshot data over live API

Initially the plan was to query HubSpot API on every page load. After consideration, this was rejected because: (1) the trial expires in 14 days, after which API calls fail; (2) recruiters viewing the demo months later would see broken errors; (3) live API adds latency on every page interaction; (4) snapshot data is reproducible — the extraction script is committed to the repo and runnable by anyone with their own token.

Theme forcing via config.toml

Streamlit defaults to user's system theme preference. For a portfolio deliverable, this is unacceptable — a recruiter on dark mode would see invisible text on light metric containers. The solution is a **.streamlit/config.toml** file that forces light theme and applies Coffra brand colors. This file is committed and applies on Streamlit Cloud automatically.

Cached data loaders

All file I/O wrapped in Streamlit's **@st.cache_data** decorator. This avoids re-reading JSON files on every page interaction. For static snapshot data this is purely a performance optimization with no correctness implications. For a live system with frequently-updating data, the cache decorator would need TTL configuration.

Honest disclosure as a feature

The Methodology page is not a footer afterthought — it is a featured page in the sidebar navigation. The data disclosure banner pattern (real / snapshot / simulated colored by type) appears on every relevant page. This signals to a senior reviewer that the author has thought about data provenance and trustworthiness as a first-class concern, not as a compliance checkbox.

Skills Demonstrated

Category	Specific Skills
Production deployment	Streamlit Cloud setup, secrets management, requirements.txt, auto-rebuild on push
Multi-page Streamlit	Page registry, sidebar navigation, shared utilities, config.toml theming
Data visualization	Plotly figure factories, brand-aligned palettes, gauge/funnel/donut/histogram patterns
API integration	HubSpot Private App API, scoped permissions, pagination, error handling
Data architecture	Snapshot vs live trade-off analysis, cached loaders, JSON-based portable artifacts
Documentation	Methodology page, provenance tables, production roadmap, inline assumptions
Brand consistency	Cross-deliverable styling (case study PDF, dashboard, email design), light theme forcing
Software hygiene	Secret management, gitignore discipline, environment templates, dependency pinning

Trade-offs accepted for portfolio scope

- **No authentication.** The dashboard is public. A real production deployment would require login.
- **No automated tests.** Manual testing only. Production system would have unit tests for data loaders and integration tests for page rendering.
- **Single-snapshot.** No scheduled refresh. Production would have a cron-based refresh or webhook trigger from HubSpot.
- **No alerting or monitoring.** If a data source breaks, the dashboard fails silently or shows empty visuals. Production would have observability.

Production roadmap

The Methodology page enumerates next steps for promoting this from portfolio to production: scheduled HubSpot refresh, GA4 integration, attribution model, multi-environment deployment, role-based access control. None of these were attempted in v1.0 — the goal was to ship a working demo with honest scope, not to build production infrastructure for a fictional brand.

Connection to P1 and Future Projects

P2 demonstrates that the P1 deliverables (lead scoring model, AI subject optimizer, HubSpot configuration) are not isolated artifacts — they integrate into a coherent operational system. Future projects in the portfolio will continue this pattern:

P3 (Customer Segmentation) will add RFM analysis and clustering to the lead scoring foundation. P4 (Recommendation Systems) will use clustering output to drive personalized product recommendations. P5 (Attribution Modeling) will close the loop by measuring channel effectiveness across the full marketing system. Each project deliberately demonstrates a non-overlapping skill cluster while building on prior infrastructure.

Contact

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Live demo: coffra-marketing-dashboard.streamlit.app · GitHub: github.com/sebikradyel1-svg